Low-field carrier mobility models for 3D electron gas (3DEG) slab

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This documentation is part of `mobilitypy` package.[1]

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Following models for low-field electron mobility (μ) calculations from different scattering mechanism are based on the Reference-[2, 3, 4, 5, 6, 7] and the references therein.

The available mobility model versions are:

- 'v1' (available in `mobilitypy` $\geq$ `v1.0.5`)

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1. Total mobility:

$$\mu = \left(\sum_i \frac{1}{\mu_i} \right)^{-1}$$

   where μ_i's are the mobility from different scattering mechanisms presented below.

2. The mobility models here allows to calculate position dependent mobility, i.e. $\mu(z)$.

1 Parameters

Parameters name	Symbol	SI Unit
Electron elementary charge	e	C
Boltzmann constant	k_B	$\mathrm{J\,K^{-1}}$
Temperature	T	K
Mass density	ρ	$\mathrm{kg\,m^{-3}}$
Lattice parameter	a_0	Å
Lattice parameter	c_0	Å
Unit cell volume	Ω_0	$\mathrm{Å}^3$
Bandgap energy	E_g	eV
Alloy-disordered scattering potential	U_0	eV
Electron effective mass	m^*	m_0
Electron rest mass	m_0	kg
Static dielectric constant	ϵ_s	ϵ_0
High frequency dielectric constant	ϵ_h	ϵ_0
Vacuum permittivity	ϵ_0	$\mathrm{C\,V^{-1}\,m^{-1}}$
Longitudinal acoustic phonon velocity	v_{LA}	$\mathrm{m\,s^{-1}}$
Transversal acoustic phonon velocity	v_{TA}	$\mathrm{m\,s^{-1}}$
Conduction band deformation potential	E_D	eV
Polar optical phonon energy	E_{POP}	eV
Electromechanical coupling coefficient	K^2	-
Elastic constants	C_{ij}	GPa
Fermi wave vector	k_F	$10^6\ \mathrm{cm^{-1}}$
Thomas-Fermi wave vector	q_{TF}	$10^6\ \mathrm{cm^{-1}}$
POP wave vector	q_0	$10^6\ \mathrm{cm^{-1}}$
Volumetric electron gas density	n_{3d}	$10^{18}\ \mathrm{cm^{-3}}$
Dislocation density	N_{DIS}	$10^8\ \mathrm{cm^{-2}}$
Dislocation occupancy	f_{DIS}	-
Mobility	μ	$\mathrm{cm^{-2}\,V^{-1}\,s^{-1}}$

Note: Proper unit conversions for all parameters have been implemented within the equations.

2 Basic theory of mobility

- **Fermi-Dirac function for fermions:**

$$f_0(E) = \frac{1}{1 + e^{\frac{E-E_F}{k_B T}}} \tag{1}$$

where E_F is the Fermi energy.

- **Boltzmann Transport Equation (BTE)**: Within relaxation time approximation:

$$f(\vec{k}) = f_0(\vec{k}) + e\mathbb{F}_i \tau(k) v_i \frac{\partial f_0}{\partial \varepsilon} \tag{2}$$

where $f(\vec{k})$ is carrier distribution function, $\tau(k)$ is the momentum relaxation time, v_i is the carriers velocity in the i-th direction in reponse of the field $\mathbb{F}_i$.

- **DOS in d-dimension:**

$$g_d(E) = \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} (E - E_C)^{\frac{d}{2}-1} \tag{3}$$

- **Total number of carriers/unit 'volume' in d-dimensional space:**

$$\begin{aligned}
n_{dd} &= \int \frac{d^d k}{(2\pi)^d} f(k) \\
&= \int_{E_C}^{\infty} f_0(E)\, g_d(E)\, dE \\
&= \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}}\, g_d(E) dE \\
&= \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} (E - E_C)^{\frac{d}{2}-1} dE \\
&= \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}} (E - E_C)^{\frac{d}{2}-1} dE \\
&= \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{\frac{(E-E_C)-(E_F-E_C)}{k_B T}}} \left(\frac{E - E_C}{k_B T}\right)^{\frac{d}{2}-1} (k_B T)^{\frac{d}{2}-1}\, dE \\
&= \left(\frac{2m^* k_B T}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{\frac{(E-E_C)-(E_F-E_C)}{k_B T}}} \left(\frac{E - E_C}{k_B T}\right)^{\frac{d}{2}-1} \frac{1}{k_B T}\, dE
\end{aligned} \tag{4}$$

- **Fermi-Dirac integral of j-th order:**

$$F_j(\zeta) = \frac{1}{\Gamma(j+1)} \int_0^\infty \frac{u^j}{1+e^{u-\zeta}}\, du$$

- Substituting $\varepsilon = \frac{E-E_C}{k_B\,T}$ and $\eta_{_F} = \frac{E_F-E_C}{k_b\,T}$

$$f_0(\varepsilon) = \frac{1}{1+e^{\varepsilon-\eta_{_F}}} \tag{5}$$

$$F_{\frac{d}{2}-1}(\eta_{_F}) = \frac{1}{\Gamma(\frac{d}{2})} \int_0^\infty \frac{\varepsilon^{\frac{d}{2}-1}}{1+e^{\varepsilon-\eta_{_F}}}\, d\varepsilon = \frac{1}{\Gamma(\frac{d}{2})} \int_0^\infty f_0(\varepsilon)\, \varepsilon^{\frac{d}{2}-1}\, d\varepsilon \tag{6}$$

$$\begin{aligned} n_{dd} &= \left(\frac{2m^*\,k_BT}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}} \frac{1}{\Gamma(\frac{d}{2})} \int_0^\infty \frac{1}{1+e^{\varepsilon-\eta_{_F}}} \varepsilon^{\frac{d}{2}-1}\, d\varepsilon \\ &= \left(\frac{2m^*\,k_BT}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}}\, F_{\frac{d}{2}-1}(\eta_{_F}) \end{aligned} \tag{7}$$

- **Current:**

$$J_i = en \underbrace{\left(-\frac{2e}{dm^*} \frac{\int_{E_C}^\infty \tau_m (E-E_C)^{d/2} \frac{\partial f_0}{\partial E} dE}{\int_{E_C}^\infty f_0\,(E-E_C)^{\frac{d}{2}-1}\, dE}\right)}_{\mu_{dd}} \mathcal{F}_i \tag{8}$$

- **Mobility in d-dimension:**

$$\mu_{dd} = -\frac{2e}{dm^*} \frac{\int_{E_C}^\infty \tau_m (E-E_C)^{d/2} \frac{\partial f_0(E)}{\partial E} dE}{\int_{E_C}^\infty f_0(E) \times (E-E_C)^{\frac{d}{2}-1}\, dE} \tag{9}$$

$$\begin{aligned} &= -\frac{2e}{dm^*} \frac{\int_{E_C}^\infty \tau_m (E-E_C)^{d/2} \frac{\partial f_0(E)}{\partial E}\, dE}{\int_{E_C}^\infty f_0(E) \times (E-E_C)^{\frac{d}{2}-1}\, dE} \\ &= -\frac{2e}{dm^*} \frac{(k_BT)^{\frac{d}{2}} \int_0^\infty \tau_m \varepsilon^{d/2}\, \frac{\partial f_0(\varepsilon)}{\partial \varepsilon} d\varepsilon}{(k_BT)^{\frac{d}{2}} \int_0^\infty f_0 \times \varepsilon^{\frac{d}{2}-1}\, d\varepsilon} \\ &= -\frac{e\tau_0}{m^*} \frac{\int_0^\infty \varepsilon^{\frac{d}{2}+p}\, \frac{\partial f_0(\varepsilon)}{\partial \varepsilon} d\varepsilon}{\frac{d}{2} \int_0^\infty f_0 \times \varepsilon^{\frac{d}{2}-1}\, d\varepsilon} \quad [\because \tau_m = \tau_0 \times \varepsilon^p] \end{aligned} \tag{10}$$

$$
\begin{aligned}
&= -\frac{e\tau_0}{m^*}\frac{-\Gamma(\frac{d}{2}+p+1)\,F_{\frac{d}{2}+p-1}(\eta_F)}{\frac{d}{2}\times\Gamma(\frac{d}{2})\times F_{\frac{d}{2}-1}(\eta_F)} \\
&= \frac{e\tau_0}{m^*}\left(\frac{\Gamma(\frac{d}{2}+p+1)}{\Gamma(\frac{d}{2}+1)}\right)\left(\frac{F_{\frac{d}{2}+p-1}(\eta_F)}{F_{\frac{d}{2}-1}(\eta_F)}\right) \qquad (11)
\end{aligned}
$$

Since, $f_0(\varepsilon)\to 0$ as $\varepsilon\to\infty$ and $\varepsilon^m\to 0$ as $\varepsilon\to 0$, using integration by parts $[\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx]$

$$
\begin{aligned}
&\int_0^\infty \varepsilon^{\frac{d}{2}+p}\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}\,d\varepsilon \\
&= [\varepsilon^{\frac{d}{2}+p}\,f_0(\varepsilon)]_0^\infty - \int_0^\infty\left(\frac{d}{2}+p\right)\times\varepsilon^{\frac{d}{2}+p-1}\,f_0(\varepsilon)\,d\varepsilon \\
&= -\left(\frac{d}{2}+p\right)\int_0^\infty \varepsilon^{\frac{d}{2}+p-1}\,f_0(\varepsilon)\,d\varepsilon \\
&= -\left(\frac{d}{2}+p\right)\,\Gamma(\frac{d}{2}+p)\,F_{\frac{d}{2}+p-1}(\eta_F) \\
&= -\Gamma(\frac{d}{2}+p+1)\,F_{\frac{d}{2}+p-1}(\eta_F) \qquad (12)
\end{aligned}
$$

- **In 3D:**

$$
\begin{aligned}
n_{3d} &= \left(\frac{2m^*\,k_BT}{\pi\hbar^2}\right)^{\frac{3}{2}}\,\frac{1}{4}\,F_{\frac{1}{2}}(\eta_F) \\
&= 2\left(\frac{m^*\,k_BT}{2\,\pi\hbar^2}\right)^{\frac{3}{2}}\,F_{\frac{1}{2}}(\eta_F) \\
&= N_C^{3d}\,F_{\frac{1}{2}}(\eta_F) \qquad (13) \\
N_C^{3d} &= 2\left(\frac{m^*\,k_BT}{2\,\pi\hbar^2}\right)^{\frac{3}{2}} \qquad (14) \\
\mu_{3d} &= \frac{e\tau_0}{m^*}\frac{\Gamma(\frac{5}{2}+p)}{\Gamma(\frac{5}{2})}\frac{F_{\frac{1}{2}+p}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)} \\
&= \frac{4e\tau_0}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(\frac{5}{2}+p)\times F_{\frac{1}{2}+p}(\eta_F) \qquad (15)
\end{aligned}
$$

3 Fermi-Dirac integral

3.1 Inverse Fermi-Dirac integral of order 1/2

The scaled Fermi energy ($\eta_F = \frac{E_F - E_C}{k_B T}$) is calculated using inverse Fermi-Dirac integral of order 1/2.

$$\begin{aligned} \because n_{3d} &= N_C^{3d}\, F_{\frac{1}{2}}(\eta_F) \Big| = N_C^{3d}\, \frac{2}{\sqrt{\pi}} \mathcal{F}_{\frac{1}{2}}(\eta_F) = N_{3d}\, \mathcal{F}_{\frac{1}{2}}(\eta_F) \\ \implies \eta_F &= F_{1/2}^{-1}\left(\frac{n_{3d}}{N_C^{3d}}\right) \Bigg| \; \eta_F = \mathcal{F}_{1/2}^{-1}\left(\frac{n_{3d}}{N_{3d}}\right) \end{aligned} \tag{16}$$

where

$$F_{\frac{1}{2}}(\eta_F) = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1 + e^{\varepsilon - \eta_F}} d\varepsilon \Bigg| \; \mathcal{F}_{\frac{1}{2}}(\eta_F) = \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1 + e^{\varepsilon - \eta_F}} d\varepsilon \tag{17}$$

$$N_C^{3d} = 2\left(\frac{m^* k_B T}{2\pi\hbar^2}\right)^{\frac{3}{2}} \Bigg| \; N_{3d} = \frac{2}{\sqrt{\pi}} N_C^{3d} = \frac{4}{\sqrt{\pi}} \left(\frac{m^* k_B T}{2\pi\hbar^2}\right)^{\frac{3}{2}} \tag{18}$$

1. **Using Joyce-Dixon approximation:**[8]

$$\begin{aligned} \eta_F &= F_{1/2}^{-1}\left(\frac{n_{3d}}{N_C^{3d}}\right) \simeq \ln\left(\frac{n_{3d}}{N_C^{3d}}\right) + \sum_{l=1}^{3} A_l \times \left(\frac{n_{3d}}{N_C^{3d}}\right)^l \\ A_l &= 0.3535534, -4.95009 \times 10^{-3}, 1.48386 \times 10^{-4} \end{aligned} \tag{19}$$

   Note that the equation here is in terms of n_{3d} and N_C^{3d}. To compare with the original derivation in Ref.[8] use this notation:
   $[\mathfrak{F} = 2\pi^{-1/2}\, F]_{publication} \equiv [F_{1/2} = 2\pi^{-1/2}\, \mathcal{F}_{1/2}]_{our}$

2. **Using minimax rational function approximation:**[9]

$$\eta_F = \mathcal{F}_{1/2}^{-1}\left(\frac{n_{3d}}{N_{3d}}\right) \simeq \text{Piecewise function [Ref.[9]]} \tag{20}$$

   Note that the equation here is in terms of n_{3d} and N_{3d}.

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$
\begin{aligned}
N_C^{3d} &= 2\left(\frac{m^*\,k_B T}{2\pi\hbar^2}\right)^{3/2} \\
&= 2\left(\frac{k_B\,m_0}{2\pi\hbar^2}\right)^{3/2} \times 10^{-6}\,\mathrm{cm}^{-3}\mathrm{K}^{-3/2} \times (m^{*'}\,T')^{3/2}\,\mathrm{K}^{3/2} \\
&= 4.83 \times (m^{*'}\,T')^{3/2} \times 10^{15}\ \mathrm{cm}^{-3} \qquad (21) \\
N_{3d} &= \frac{2}{\sqrt{\pi}} N_C^{3d} = \frac{2}{\sqrt{\pi}} \times 4.83 \times (m^{*'}\,T')^{3/2} \times 10^{15}\ \mathrm{cm}^{-3} \qquad (22) \\
&= 5.45 \times (m^{*'}\,T')^{3/2} \times 10^{15}\ \mathrm{cm}^{-3} \qquad (23) \\
\frac{n_{3d}}{N_C^{3d}} &= \frac{n'_{3d} \times 10^{18}}{4.83 \times (m^{*'}\,T')^{3/2} \times 10^{15}} = 207.07 \times \frac{n'_{3d}}{(m^{*'}\,T')^{3/2}} \qquad (24) \\
\frac{n_{3d}}{N_{3d}} &= \frac{n'_{3d} \times 10^{18}}{5.45 \times (m^{*'}\,T')^{3/2} \times 10^{15}} = 183.51 \times \frac{n'_{3d}}{(m^{*'}\,T')^{3/2}} \qquad (25)
\end{aligned}
$$

3.2 Fermi-Dirac integrals

1. η_F is calculated using expression from Sec.3.1.
2. **Fermi-Dirac integral of order 0: (analytical solution)**

$$F_0(\eta_F) = \frac{1}{\Gamma(1)} \int_0^\infty \frac{1}{1+e^{\varepsilon-\eta_F}} d\varepsilon = \ln[1+\exp(\eta_F)]$$

3. **Fermi-Dirac integral of order 1/2:**

$$F_{\frac{1}{2}}(\eta_F) = \frac{1}{\Gamma(\frac{1}{2}+1)} \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}} d\varepsilon = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}} d\varepsilon$$

4. **Fermi-Dirac integral of order 1:**

$$
\begin{aligned}
F_1(\eta_F) &= \frac{1}{\Gamma(2)} \int_0^\infty \frac{\varepsilon}{1+e^{\varepsilon-\eta_F}} d\varepsilon = \int_0^\infty \frac{\varepsilon}{1+e^{\varepsilon-\eta_F}} d\varepsilon \qquad (26) \\
F_1(\eta_F) &= -\mathrm{Li}_2(-e^{\eta_F}) \qquad (27) \\
\mathrm{Li}_2(z) &= -\int_0^z \frac{\ln(1-t)}{t} dt = \texttt{scipy.spence(1-z)} \\
\Longrightarrow \mathrm{Li}_2(-e^{\eta_F}) &= \texttt{scipy.spence(1+e^eta_f)} \\
\Longrightarrow F_1(\eta_F) &= (-1) \times \texttt{scipy.spence(1+e^eta_f)} \qquad (28)
\end{aligned}
$$

The integration can be calculated numerically using Eqn. 26 (implemented using `scipy.quad`) or using dilogarithm formalism of complete Fermi-Dirac integral of order-1, Eqn. 27 (implemented as `(-1) * scipy.spence(1+e^eta_f)`). Both are implemented in our package.

4 Scattering mechanisms

$$k_F = \left(3\pi^2 n_{3D}\right)^{\frac{1}{3}} \tag{29}$$

$$\lambda_{TF} = q_{TF}^{-1} = \sqrt{\frac{2\epsilon\varepsilon_F}{3e^2 n_{3D}}} = \sqrt{\frac{2\epsilon}{3e^2 n_{3D}}\frac{\hbar^2 k_F^2}{2\,m^*}} = \frac{\hbar k_F}{e}\sqrt{\frac{\epsilon}{3n_{3D}m^*}}$$

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2\,m^*} \tag{30}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$k_F = \left(3\pi^2 n_{3D}\right)^{\frac{1}{3}} = 3.094 \times n_{3D}'^{\frac{1}{3}}\ 10^6\,\mathrm{cm}^{-1} \tag{31}$$

$$\lambda_{TF} = \frac{\hbar\left(3\pi^2 n_{3D}\right)^{\frac{1}{3}}}{e}\sqrt{\frac{\epsilon_s}{3n_{3D}m^*}} = \frac{\hbar\,\pi^{\frac{2}{3}}\epsilon_0^{\frac{1}{2}}}{3^{\frac{1}{6}}\,e\,m_0^{\frac{1}{2}}} \times \left(\frac{\epsilon_s}{n_{3D}^{\frac{1}{3}}\,m'^*}\right)^{\frac{1}{2}}$$

$$= 0.037 \times \left(\frac{\epsilon_s}{n_{3D}^{\frac{1}{3}}\,m^{*\prime}}\right)^{\frac{1}{2}}\ 10^{-6}\,\mathrm{cm} \tag{32}$$

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2\,m^*} = \frac{\hbar^2 \times (10^6)^2 \times 10^4}{2\,m_0 \times e'}\frac{k_F'^2}{m'^*} = 0.00038 \times \frac{k_F'^2}{m'^*}\ \mathrm{eV} \tag{33}$$

4.1 Structural related scattering

4.1.1 Alloy-disorder scattering

The implementation is based on Refs.[3, 4]. For alloy disorder mediated scattering $p = -\frac{1}{2}$. The momentum relaxation time:

$$\tau_{0,AD} = \frac{\hbar}{2\sqrt{\pi}}\frac{k_B\,T}{U_0^2\,\Omega_0\,x\,(1-x)\,N_C^{3d}} \tag{34}$$

Mobility expression:

$$\begin{aligned}
\mu_{3d,AD} &= \frac{4e\,\tau_{0,AD}}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(2)\times F_0(\eta_F)\ \text{[From Eqn.15]} \\
&= \frac{2e\hbar}{3\pi}\frac{k_B\,T}{m^*\,U_0^2\,\Omega_0\,x\,(1-x)}\frac{1}{N_C^{3d}\,F_{\frac{1}{2}}(\eta_F)}F_0(\eta_F) \\
&= \frac{2e\hbar}{3\pi}\frac{k_B\,T}{m^*\,U_0^2\,\Omega_0\,x\,(1-x)\,n_{3d}}F_0(\eta_F)\ \text{[From Eqn.13]} \qquad (35) \\
& \qquad (36)
\end{aligned}$$

Re-writing as function position, z:

$$\mu_{3d,AD}(z) = \frac{2e\hbar}{3\pi}\frac{k_B\,T}{m^*(z)\,U_0(z)^2\,\Omega_0(z)\,x(z)\,(1-x(z))\,n_{3d}(z)}F_0(\eta_F(z)) \qquad (37)$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu_{3d,AD}(z) &= \frac{2e\hbar}{3\pi}\frac{k_B\,T}{m^*(z)\,U_0(z)^2\,\Omega_0(z)\,x(z)\,(1-x(z))\,n_{3d}(z)}F_0(\eta_F(z)) \\
&= \frac{2e\hbar\,k_B\,10^4}{3\pi m_0\,e^2\,(10^{-8})^3\,10^{18}}\frac{T}{m^*(z)\,U_0(z)^2\,\Omega_0(z)\,x(z)\,(1-x(z))\,n_{3d}(z)} \\
&\qquad \times F_0(\eta_F(z))\ \mathrm{cm^2\,V^{-1}\,s^{-1}\,K^{-1}} \\
&= 21.16991\times\frac{T'}{m^{*\prime}(z)\,U_0'(z)^2\,\Omega_0'(z)\,x(z)\,(1-x(z))\,n_{3d}'(z)} \\
&\qquad \times F_0(\eta_F(z))\ \mathrm{cm^2\,V^{-1}\,s^{-1}} \qquad (38)
\end{aligned}$$

4.1.2 Dislocation momentum relaxation rate

4.1.2.1 Scattering from line charge of threading dislocation core

The implementation is based on Refs.[10, 5, 7, 6]. Note that, momentum relaxation time in this case can not be write as polynomial function of energy.

The momentum relaxation time:

$$\begin{aligned}
\frac{1}{\tau_{DIS}^{3d}} &= \frac{m^* e^4 n_{DIS} f_{DIS}^2}{c_0^2 \epsilon_s^2 \hbar^3} \times \frac{\lambda_{TF}^4}{(1+4\,k_F^2 \lambda_{TF}^2)^{3/2}} && (39)\\
&= \frac{m^* e^4 n_{DIS} f_{DIS}^2}{c_0^2 \epsilon_s^2 \hbar^3} \times \frac{\hbar^4 k_F^4 \epsilon_s^2}{9\,e^4\,n_{3D}^2\,m^{*2}} \times \frac{1}{(1+4\,k_F^2 \lambda_{TF}^2)^{3/2}}\\
&= \frac{n_{DIS} f_{DIS}^2 \hbar}{m^* c_0^2} \times \frac{k_F^4}{(3\,n_{3D})^2} \times \frac{1}{(1+4\,k_F^2 \lambda_{TF}^2)^{3/2}}\\
&= \frac{n_{DIS} f_{DIS}^2 \hbar}{m^* c_0^2} \times \frac{(3\pi^2 n_{3D})^{\frac{4}{3}}}{(3\,n_{3D})^2} \times \frac{1}{(1+4\,k_F^2 \lambda_{TF}^2)^{3/2}}\\
&= \frac{n_{DIS} f_{DIS}^2 \hbar}{m^* c_0^2} \times \frac{\pi^{8/3}}{(3\,n_{3D})^{2/3}} \times \frac{1}{(1+4\,k_F^2 \lambda_{TF}^2)^{3/2}} && (40)
\end{aligned}$$

Mobility expression:

$$\begin{aligned}
\mu_{DIS}^{3d} &= \frac{e}{m^*}\tau_{DIS}^{3d} = \frac{e}{m^*}\frac{m^* c_0^2}{n_{DIS} f_{DIS}^2 \hbar}\frac{(3\,n_{3D})^{2/3}}{\pi^{8/3}}(1+4\,k_F^2 \lambda_{TF}^2)^{3/2}\\
k_F &= \left(3\pi^2 n_{3D}\right)^{\frac{1}{3}} && (41)\\
\lambda_{TF} &= \frac{\hbar k_F}{e}\sqrt{\frac{\epsilon_s}{3 n_{3D} m^*}} \implies \lambda_{TF}^4 = \frac{\hbar^4 k_F^4 \epsilon_s^2}{9\,e^4\,n_{3D}^2\,m^{*2}} && (42)\\
&= \frac{3^{2/3}}{\pi^{8/3}}\frac{e}{\hbar}\frac{c_0^2\,n_{3D}^{2/3}}{n_{DIS} f_{DIS}^2}(1+4\,k_F^2 \lambda_{TF}^2)^{3/2} && (43)\\
4\,k_F^2 \lambda_{TF}^2 &= \frac{4}{3}\frac{\hbar^2\,k_F^4 \epsilon_s}{e^2\,n_{3D}\,m^*} = \frac{4\pi^{\frac{8}{3}}}{3}\frac{\hbar^2\,\epsilon_s\,(3\,n_{3D})^{4/3}}{e^2\,n_{3D}\,m^*} = 4\pi^{\frac{8}{3}}\frac{(3\,n_{3D})^{1/3}\,\hbar^2\,\epsilon_s}{e^2\,m^*} && (44)
\end{aligned}$$

Re-writing as function position, z:

$$\begin{aligned}
\mu_{DIS}^{3d} &= \frac{3^{2/3}}{\pi^{8/3}}\frac{e}{\hbar}\frac{c_0(z)^2\,n_{3D}(z)^{2/3}}{n_{DIS} f_{DIS}^2}(1+4\,k_F^2 \lambda_{TF}^2)^{3/2} && (45)\\
4\,k_F^2 \lambda_{TF}^2 &= 4\pi^{8/3} \times 3^{\frac{1}{3}} \times \frac{\hbar^2}{e^2} \times \frac{n_{3D}(z)^{1/3}\,\epsilon_s(z)}{m^*(z)} && (46)
\end{aligned}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu^{3d}_{DIS} &= \frac{3^{2/3}}{\pi^{8/3}}\frac{e}{\hbar}\frac{c_0(z)^2\, n_{3D}(z)^{2/3}}{n_{DIS} f^2_{DIS}}(1+4\,k_F^2\lambda^2_{TF})^{3/2} \\
&= 149.273 \times \frac{c_0'(z)^2\, n'_{3D}(z)^{2/3}}{n'_{DIS} f'^2_{DIS}}(1+4\,k_F^2\lambda^2_{TF})^{3/2}\ \mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1} \\
4\,k_F^2\lambda^2_{TF} &= 4\pi^{8/3}\times 3^{\frac{1}{3}} \times \frac{\hbar^2\epsilon_0}{e^2\, m_0} \times \frac{n_{3D}(z)^{1/3}\,\epsilon_s'(z)}{m^{*\prime}(z)} \\
&= 0.0514 \times \frac{n'_{3D}(z)^{1/3}\,\epsilon_s'(z)}{m^{*\prime}(z)} \qquad (47)
\end{aligned}$$

4.1.2.2 Classical-quantum scattering time ratio for charged dislocation scattering

The implementation is based on Refs.[10, 7, 6].

$$\frac{1}{\tau^q_{dis}}(k) = \frac{m^*\, e^4\, n_{dis} f^2}{c_0^2\,\epsilon_s^2\,\hbar^3}\frac{\lambda^4}{(1+4\,k^2\lambda^2)^{\frac{3}{2}}}(1+2\,k^2\lambda^2) \qquad (48)$$

$$\frac{1}{\tau^c_{dis}}(k) = \frac{m^*\, e^4\, n_{dis}\, f^2}{c_0^2\,\epsilon_s^2\,\hbar^3}\frac{\lambda^4}{(1+4\,k^2\lambda^2)^{\frac{3}{2}}} \qquad (49)$$

$$\begin{aligned}
\Longrightarrow \left.\frac{\tau^c_{dis}}{\tau^q_{dis}}\right|_{dis} &= 1+2\,k^2\lambda^2 \Rightarrow 1+2\,k_F^2\lambda^2_{TF} \\
&= 1+0.0257\times\frac{n'^{1/3}_{3D}\,\epsilon_s'}{m^{*\prime}} \qquad (50)
\end{aligned}$$

4.2 Phonon related scattering

4.2.1 Deformation potential scattering

The implementation is based on Refs.[3]. For deformation potential acoustic phonon mediated scattering $p=-\frac{1}{2}$. The momentum relaxation time:

$$\tau_{0,DP} = \frac{\hbar}{2\sqrt{\pi}}\frac{\rho\, v^2_{LA}}{E_D^2\, N_C^{3d}} \qquad (51)$$

Mobility expression:

$$\begin{aligned}
\mu_{3d,DP} &= \frac{4e\,\tau_{0,DP}}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(2)\times F_0(\eta_F)\ \text{[From Eqn.15]} \\
&= \frac{4e\hbar}{6\pi\,m^*}\frac{\rho\,v_{LA}^2}{E_D^2}\frac{1}{N_C^{3d}\,F_{\frac{1}{2}}(\eta_F)}F_0(\eta_F) \\
&= \frac{2e\hbar}{3\pi\,m^*}\frac{\rho\,v_{LA}^2}{E_D^2\,n_{3d}}F_0(\eta_F)\ \text{[From Eqn.13]} \qquad (52)
\end{aligned}$$

Re-writing as function position, z:

$$\mu_{3d,DP}(z) = \frac{2e\hbar}{3\pi}\frac{\rho(z)\,v_{LA}(z)^2}{m^*(z)\,E_D(z)^2\,n_{3d}(z)}F_0(\eta_F(z)) \qquad (53)$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu_{3d,DP}(z) &= \frac{2e\hbar}{3\pi}\frac{\rho(z)\,v_{LA}(z)^2}{m^*(z)\,E_D(z)^2\,n_{3d}(z)}F_0(\eta_F(z)) \\
&= \frac{2e\hbar\,10^{-2}}{3\pi m_0\,e^2\,10^{18}}\frac{\rho'(z)\,v'_{LA}(z)^2}{m^{*\prime}(z)\,E'_D(z)^2\,n'_{3d}(z)}F_0(\eta_F(z))\ \text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1} \\
&= 1.5333\times 10^{-6}\frac{\rho'(z)\,v'_{LA}(z)^2}{m^{*\prime}(z)\,E'_D(z)^2\,n'_{3d}(z)}F_0(\eta_F(z))\ \text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1}
\end{aligned} \qquad (54)$$

4.2.2 Piezoelectric phonon scattering

The implementation is based on Refs.[3, 2]. Note following [2] we use K^2 term in our below derivation which is related to the terms in Ref.[3] as:

$$\rho v_s^2 \equiv C^*; K^2 \equiv \langle\frac{h_{pz}^2}{C^*\epsilon_s}\rangle$$

For piezoelectric phonon scattering $p = +\frac{1}{2}$. The momentum relaxation time:

$$\tau_{0,PE} = \frac{2m^*\epsilon_s\,k_BT}{\sqrt{\pi}\,\hbar\,e^2\,K^2}\frac{1}{N_C^{3d}} \qquad (55)$$

(56)

Mobility expression:

$$\begin{aligned}
\mu_{3d,PE} &= \frac{4e\,\tau_{0,PE}}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(3)\times F_1(\eta_F)\ \text{[From Eqn.15]} \\
&= \frac{16\epsilon_s\,k_BT}{3\pi\hbar eK^2}\frac{1}{N_C^{3d}\,F_{\frac{1}{2}}(\eta_F)}F_1(\eta_F) \\
&= \frac{16\,k_BT}{3\pi\hbar e}\frac{\epsilon_s}{K^2\,n_{3d}}F_1(\eta_F)\ \text{[From Eqn.13]} \qquad (57)
\end{aligned}$$

Re-writing as function position, z:

$$\mu_{3d,PE}(z) = \frac{16\,k_BT}{3\pi\hbar e}\frac{\epsilon_s(z)}{K^2(z)\,n_{3d}(z)}F_1(\eta_F(z)) \qquad (58)$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu_{3d,PE}(z) &= \frac{16\,k_B}{3\pi\hbar e}\frac{\epsilon_s(z)\,T}{K^2(z)\,n_{3d}(z)}F_1(\eta_F(z)) \\
&= \frac{16\,k_B\epsilon_0}{3\pi\hbar e}\frac{\epsilon_s'(z)\,T}{K^{2\prime}(z)\,n_{3d}'(z)\times 10^{18}}F_1(\eta_F(z))\ \mathrm{cm^2\,V^{-1}\,s^{-1}\,K^{-1}} \\
&= 0.12283\times\frac{\epsilon_s'(z)\,T'}{K^{2\prime}(z)\,n_{3d}'(z)}F_1(\eta_F(z))\ \mathrm{cm^2\,V^{-1}\,s^{-1}} \qquad (59)
\end{aligned}$$

4.2.3 Polar optical phonon scattering

The implementation is based on Refs.[10, 3, 4]. Note that, since POP scatterings are inelastic in nature and there is no simple energy dependence unlike mechanisms discussed above, there is no general analytical expression for the POP carrier mobility. However, assuming POP energies satisfy $\hbar\omega_{LO} >> k_BT$ (results from BTE when assuming a much higher emission rate than the absorption rate) the mobility is:

$$\begin{aligned}
\mu_{0,POP} &= \frac{4\pi\epsilon^*\hbar}{e\,m^*q_0}\frac{1}{N_{BE}} \\
&= \frac{2^{\frac{3}{2}}\pi\epsilon^*\hbar^2}{e\,m^{*\frac{3}{2}}E_{pop}^{\frac{1}{2}}}\times\left(e^{\frac{E_{pop}}{k_BT}}-1\right) \qquad (60)
\end{aligned}$$

$$\frac{1}{\epsilon^*} = \frac{1}{\epsilon_h}-\frac{1}{\epsilon_s} \qquad (61)$$

$$\begin{aligned}
\hbar\omega_{pop} &= \frac{\hbar^2\, q_0^2}{2m^*} \implies q_0 = \sqrt{\frac{2m^*\, E_{POP}}{\hbar^2}} \\
N_{BE} &= \frac{1}{e^{\frac{\hbar\omega_{pop}}{k_B T}} - 1} \text{ [Bose-Einstein distribution]}
\end{aligned} \tag{62}$$

Re-writing as function position, z:

$$\mu_{3d,POP}(z) = \frac{2^{\frac{3}{2}}\pi\hbar^2}{e} \frac{\epsilon^*(z)}{m^*(z)^{\frac{3}{2}} E_{pop}(z)^{\frac{1}{2}}} \times \left(e^{\frac{E_{pop}(z)}{k_B T}} - 1\right) \tag{63}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu_{3d,POP}(z) &= \frac{2^{\frac{3}{2}}\pi\hbar^2}{e} \frac{\epsilon^*(z)}{m^*(z)^{\frac{3}{2}} E_{pop}(z)^{\frac{1}{2}}} \times \left(e^{\frac{E_{pop}(z)}{k_B T}} - 1\right) \\
&= \frac{2^{\frac{3}{2}}\pi\hbar^2\epsilon_0 \times 10^4}{e \times m_0^{3/2} \times e^{1/2}} \frac{\epsilon^{*\prime}(z)}{m^{*\prime}(z)^{\frac{3}{2}} E'_{pop}(z)^{\frac{1}{2}}} \times \left(e^{\frac{e \times E'_{pop}(z)}{k_B T'}} - 1\right) \\
&\qquad \text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1} \\
&= 0.15693 \times \frac{\epsilon^{*\prime}(z)}{m^{*\prime}(z)^{\frac{3}{2}} E'_{pop}(z)^{\frac{1}{2}}} \times \left(e^{11604.518 \times \frac{E'_{pop}(z)}{T'}} - 1\right) \\
&\qquad \text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1}
\end{aligned} \tag{64}$$

5 General 3DEG properties

1. Fermi wave vector

$$k_F = \left(3\pi^2 n_{3D}\right)^{\frac{1}{3}} = 3.09 \times n'^{\frac{1}{3}}_{3D} \;\; [10^6\,\text{cm}^{-1}] \tag{65}$$

2. Fermi energy (parabolic dispersion)

$$\begin{aligned}
E_F &= \frac{\hbar^2\, k_F^2}{2\, m^*} = \frac{\hbar^2}{2\, m_0} \frac{k'^2_F}{m^{*\prime}} \times (10^6\,\text{cm}^{-1})^2 \\
&= \frac{\hbar^2}{2\, m_0\, e'} 10^{12} \times \frac{k'^2_F}{m^{*\prime}} = 0.00038 \times \frac{k'^2_F}{m^{*\prime}} \text{ [eV]}
\end{aligned} \tag{66}$$

3. Thomas-Fermi screening wave vector

$$
\begin{aligned}
q_{TF} = 1/\lambda_{TF} &= \frac{e}{\hbar k_F}\sqrt{\frac{3n_{3D}m^*}{\epsilon_s}} = \frac{e}{\hbar\,(3\pi^2 n_{3D})^{\frac{1}{3}}}\sqrt{\frac{3n_{3D}m^*}{\epsilon_s}} \\
&= \frac{e\,m_0^{\frac{1}{2}}\,3^{\frac{1}{6}}}{\hbar\,\epsilon_0^{\frac{1}{2}}\pi^{\frac{2}{3}}}\frac{m^{*\prime\frac{1}{2}}n_{3D}^{\prime\frac{1}{6}}}{\epsilon_s^{\prime\frac{1}{2}}} \times 10^3\,\mathrm{cm}^{-1} \\
&= 27.28 \times \sqrt{\frac{m^{*\prime}n_{3D}^{\prime\frac{1}{3}}}{\epsilon_s}}\ \ [10^6\,\mathrm{cm}^{-1}] \qquad (67)
\end{aligned}
$$

4. Classical-quantum scattering time ratio for charged dislocation

$$
\left.\frac{\tau^c_{dis}}{\tau^q_{dis}}\right|_{dis} = 1 + 2\,k_F^2\lambda_{TF}^2 = 1 + 0.0257 \times \frac{n_{3D}^{\prime 1/3}\,\epsilon_s^{\prime}}{m^{*\prime}} \qquad (68)
$$

5. POP wave vector

$$
\begin{aligned}
q_0 &= \sqrt{\frac{2m^*\,E_{POP}}{\hbar^2}} = \sqrt{\frac{2\,m_0\,e'}{\hbar^2}}\sqrt{m^{*\prime}E'_{POP}} \\
&= 51.23 \times \sqrt{m^{*\prime}E'_{POP}}\ \ [10^6\,\mathrm{cm}^{-1}] \qquad (69)
\end{aligned}
$$

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